Smoothing polyhedral Lagrangian surfaces in $\mathbb{R}^4$ with 4 faces at each vertex

Abstract

We prove that an embedded polyhedral Lagrangian surface $K \subset \mathbb{R}^4$ for which exactly four 2–faces meet at every vertex admits a Lagrangian smoothing in the sense of the problem statement: there exists a one–parameter family K_t of smooth embedded Lagrangian surfaces, $t \in (0, 1]$, which are mutually Hamiltonian isotopic and which converge (and extend) to K as $t \to 0$ through a topological isotopy.

1 Statement

We work in $(\mathbb{R}^4, \omega_0)$ with standard symplectic form

$$\omega_0 = dx_1 \wedge dy_1 + dx_2 \wedge dy_2,$$

and standard Liouville 1–form

$$\lambda_0 = y_1\, dx_1 + y_2\, dx_2, \qquad d\lambda_0 = \omega_0.$$

We identify $\mathbb{R}^4 \cong T^*\mathbb{R}^2$ via $(x_1, x_2, y_1, y_2) = (q_1, q_2, p_1, p_2)$.

Definition 1 (Polyhedral Lagrangian surface)**.** A *polyhedral Lagrangian surface* $K \subset \mathbb{R}^4$ is a finite 2–dimensional polyhedral complex embedded as a topological 2–manifold in $\mathbb{R}^4$, such that every 2–face is contained in an *affine* Lagrangian 2–plane (equivalently, ω_0 restricts to 0 on the affine span of each face).

Definition 2 (Lagrangian smoothing)**.** A *Lagrangian smoothing* of such a K is a family $\{K_t\}_{t\in(0,1]}$ of smooth embedded Lagrangian submanifolds of $\mathbb{R}^4$ which are mutually Hamiltonian isotopic for $t > 0$, and which extends to a continuous family $\{K_t\}_{t\in[0,1]}$ of embedded topological surfaces with $K_0 = K$ (a topological isotopy).

The question asks whether the following hypothesis forces existence of a Lagrangian smoothing:

Hypothesis: exactly 4 2–faces meet at every vertex of K.

Our main result is an affirmative answer.

Theorem 1. *Let $K \subset (\mathbb{R}^4, \omega_0)$ be a polyhedral Lagrangian surface. Assume that exactly four 2–faces meet at every vertex of K. Then K admits a Lagrangian smoothing in the sense above.*

Remark 1. The proof is constructive and local: each edge and vertex singularity is replaced by a smooth Lagrangian patch obtained from a C^1 piecewise–quadratic generating function by mollification and cutoff. The patches can be chosen in pairwise disjoint neighborhoods, so their Hamiltonian deformations commute (up to composition), producing a global Hamiltonian family for $t > 0$ and a C^0–limit as $t \to 0$.

2 Linear algebra: Lagrangian planes as graphs of symmetric maps

Fix a Lagrangian 2–plane $L_0 \subset \mathbb{R}^4$. After a linear symplectomorphism we may assume $L_0 = \{(q, p) \in T^*\mathbb{R}^2 : p = 0\}$.

Lemma 1 (Graphs)**.** *A 2–plane $L \subset T^*\mathbb{R}^2$ is Lagrangian and transverse to the zero section $\{p = 0\}$ if and only if*

$$L = \{(q, Aq) : q \in \mathbb{R}^2\}$$

for a unique symmetric linear map $A : \mathbb{R}^2 \to (\mathbb{R}^2)^* \cong \mathbb{R}^2$ (equivalently, a unique symmetric 2×2 matrix).

Proof. If L is transverse to $\{p = 0\}$, the projection $\pi(q, p) = q$ restricts to a linear isomorphism $L \to \mathbb{R}^2$, so L is the graph of a linear map $A : \mathbb{R}^2 \to \mathbb{R}^2$, i.e. $p = Aq$. The restriction of $\omega_0 = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$ to the graph is

$$\omega_0|_{\mathrm{graph}(A)} = dq \wedge d(Aq) = dq \wedge (A\, dq),$$

which vanishes if and only if A is symmetric. Conversely, if A is symmetric, the graph is Lagrangian. Uniqueness is clear. □

Lemma 2 (Intersection lines)**.** *Let $L_A = \mathrm{graph}(A)$ and $L_B = \mathrm{graph}(B)$ be two Lagrangian planes transverse to $\{p = 0\}$. Then $L_A \cap L_B$ is nontrivial if and only if $\ker(A - B) \neq \{0\}$, and in that case*

$$L_A \cap L_B = \{(q, Aq) : q \in \ker(A - B)\}.$$

In particular, if $L_A \cap L_B$ is 1–dimensional, then $\ker(A - B)$ is a line in $\mathbb{R}^2$.

Proof. A point lies in both graphs iff $Aq = Bq$, i.e. $(A - B)q = 0$. The remaining statements are immediate. □

3 Local normal form near a polyhedral edge and vertex

Because each face is planar, the local model near a vertex is literally a cone (a union of planar sectors through the vertex). The same holds near an edge away from its endpoints.

3.1 Edge model (two faces)

Let two Lagrangian faces meet along a line segment (an edge). In a sufficiently small neighborhood away from endpoints, the surface is locally the union of two planar sectors in two Lagrangian planes meeting along their common intersection line.

Proposition 1 (Two-face corner as dF)**.** *Let $C \subset T^*\mathbb{R}^2$ be the union of two closed planar sectors $C^\pm \subset L^\pm$, where $L^\pm$ are Lagrangian planes through 0, $L^+ \neq L^-$, and $L^+ \cap L^-$ is a line. Assume moreover that $L^\pm$ are transverse to $\{p = 0\}$. Then there exist symmetric matrices $A_\pm$ and a line $\ell \subset \mathbb{R}^2$ such that, writing $\mathbb{R}^2 = \mathbb{R}^2_+ \cup \ell \cup \mathbb{R}^2_-$ as the two closed half–planes bounded by ℓ, the set C equals the graph of dF where*

$$F(q) = \tfrac{1}{2} q^T A_+ q \quad \text{for } q \in \mathbb{R}^2_+, \qquad F(q) = \tfrac{1}{2} q^T A_- q \quad \text{for } q \in \mathbb{R}^2_-,$$

and F is C^1 across ℓ.

Proof. By Lemma 1, $L^\pm =$

3.2 Vertex model (four faces)

Now assume that at a vertex exactly four faces meet. Translating the vertex to the origin, the neighborhood is the cone on a quadrilateral in the link sphere, i.e. a cyclic union of four planar sectors in four Lagrangian planes meeting pairwise along four edge lines.

Proposition 2 (Four-face vertex as dF)**.** *Let $C \subset \mathbb{R}^4$ be the star of a vertex in a polyhedral Lagrangian surface, translated so the vertex is at 0, and assume C is the union of exactly four planar sectors $C_i \subset P_i$ ($i \in \{1,2,3,4\}$) such that:*

- *each P_i is a Lagrangian plane through 0,*
- *consecutive planes meet in a line: $P_i \cap P_{i+1} = \ell_i$ (indices mod 4),*
- *the union C is a topological cone on a 4–gon (equivalently, its link is a topological circle subdivided into four edges).*

Then there exists a linear symplectomorphism Φ of $(\mathbb{R}^4, \omega_0)$ and a C^1 function $F : \mathbb{R}^2 \to \mathbb{R}$ which is quadratic on each of four closed angular sectors in $\mathbb{R}^2$ and satisfies

$$\Phi(C) = \mathrm{graph}(dF) \cap U$$

for some neighborhood U of 0 in $T^*\mathbb{R}^2$.

Proof. Choose a Lagrangian plane L_0 transverse to each P_i.[1] Pick a linear symplectomorphism Φ sending L_0 to the zero section $\{p = 0\}$, so we work in $T^*\mathbb{R}^2$.

Then each $\Phi(P_i)$ is a Lagrangian plane transverse to $\{p = 0\}$, hence $\Phi(P_i) = \mathrm{graph}(A_i)$ for symmetric A_i (Lemma 1).

The intersection line $\ell_i = P_i \cap P_{i+1}$ is mapped to $\Phi(\ell_i) \subset \Phi(P_i) \cap \Phi(P_{i+1})$. By Lemma 2, its projection to the base $\mathbb{R}^2$ is the line

$$r_i := \ker(A_i - A_{i+1}) \subset \mathbb{R}^2.$$

Since C is a cone on a 4–gon, the four lines r_i are distinct and occur in cyclic order, hence partition a small disk around 0 in $\mathbb{R}^2$ into four (closed) angular sectors Σ_i with boundary rays contained in $r_{i-1} \cup r_i$.

Define

$$F(q) = \tfrac{1}{2} q^T A_i q \quad \text{for } q \in \Sigma_i.$$

Along a boundary ray contained in $r_i = \ker(A_i - A_{i+1})$, we have $A_i q = A_{i+1} q$, hence dF agrees from the two sides on the boundary ray. Moreover, since $q^T(A_i - A_{i+1})q = q^T(A_i q - A_{i+1} q) = 0$ on r_i, the *values* of F also agree along that ray (and $F(0) = 0$), so F is globally continuous and C^1.

Finally, by construction $\mathrm{graph}(dF)$ agrees with $\mathrm{graph}(A_i) = \Phi(P_i)$ over Σ_i, so $\mathrm{graph}(dF)$ coincides with $\Phi(C)$ in a neighborhood of 0, proving the claim. □

[1] Existence: the Lagrangian Grassmannian $\mathrm{Lag}(2,4) \cong U(2)/O(2)$ is a 3–manifold. For fixed P_i, the subset of L_0 not transverse to P_i is a proper closed subset of codimension ≥ 1. Since there are finitely many P_i, their union does not exhaust $\mathrm{Lag}(2,4)$, so a transverse L_0 exists.

4 Smoothing by mollification and cutoff

4.1 Mollifiers

Fix a standard nonnegative mollifier $\rho \in C_c^\infty(\mathbb{R}^2)$ with $\int_{\mathbb{R}^2} \rho = 1$ and set $\rho_\varepsilon(q) = \varepsilon^{-2}\rho(q/\varepsilon)$.

Lemma 3 (Convolution smooths in C^1)**.** *If $F : \mathbb{R}^2 \to \mathbb{R}$ is C^1 with locally Lipschitz derivative (in particular, if F is piecewise quadratic and C^1), then $F_\varepsilon := \rho_\varepsilon * F$ is smooth and $F_\varepsilon \to F$ in C^1 on compact sets as $\varepsilon \to 0$.*

Proof. This is standard: $F \in C^1$ with locally Lipschitz derivative implies dF is locally uniformly continuous. Convolution commutes with derivatives and $dF_\varepsilon = \rho_\varepsilon * dF$, which converges uniformly on compact sets to dF as $\varepsilon \to 0$. □

4.2 Cutoff to keep the surface fixed away from the singular locus

Let $\chi \in C_c^\infty(\mathbb{R}^2)$ be a cutoff with $\chi \equiv 1$ on $B_1(0)$ and $\chi \equiv 0$ outside $B_2(0)$. For a scale parameter $\delta > 0$ set $\chi_\delta(q) = \chi(q/\delta)$.

Given a piecewise–quadratic C^1 function F on a neighborhood of 0, define

$$\widetilde{F}_{\varepsilon,\delta}(q) := \chi_\delta(q)\,(\rho_\varepsilon * F)(q) + \big(1 - \chi_\delta(q)\big)\,F(q).$$

Then $\widetilde{F}_{\varepsilon,\delta}$ is smooth (since $\rho_\varepsilon * F$ is smooth and χ_δ is smooth), and $\widetilde{F}_{\varepsilon,\delta} = F$ outside $B_{2\delta}(0)$.

Lemma 4 (Graphs of exact 1–forms are Lagrangian)**.** *For any C^2 function G on an open set in $\mathbb{R}^2$, the graph $\Gamma_{dG} = \{(q, dG(q))\}$ is a smooth Lagrangian submanifold of $T^*\mathbb{R}^2$.*

Proof. This is standard: $p = dG$ implies $\lambda_0|_{\Gamma_{dG}} = p \cdot dq = dG$, hence $\omega_0|_{\Gamma_{dG}} = d\lambda_0|_{\Gamma_{dG}} = d(dG) = 0$. □

4.3 Hamiltonian isotopies between graphs

We now record the basic fact that varying a generating function gives a Hamiltonian isotopy of graphs.

Proposition 3 (Hamiltonian realizing variation of a generating function)**.** *Let G_s be a smooth one-parameter family of smooth functions on $\mathbb{R}^2$ such that G_s is constant (or has compactly supported variation) outside a fixed compact set. Then the family of graphs Γ_{dG_s} is a Hamiltonian isotopy in $T^*\mathbb{R}^2$. In fact, the time-dependent Hamiltonian*

$$H_s(q, p) := -\partial_s G_s(q)$$

generates a Hamiltonian vector field whose flow sends $\Gamma_{dG_{s_0}}$ to $\Gamma_{dG_{s_1}}$.

Proof. Compute X_{H_s} from $\omega_0(\cdot, X_{H_s}) = dH_s$: since H_s depends only on q, we have $dH_s = -d(\partial_s G_s)$ and therefore

$$X_{H_s} = \sum_{j=1}^{2} \frac{\partial H_s}{\partial q_j}\,\frac{\partial}{\partial p_j} = -\sum_{j=1}^{2} \frac{\partial^2 G_s}{\partial s\,\partial q_j}\,\frac{\partial}{\partial p_j},$$

a vertical vector field. Thus the Hamiltonian flow fixes q and translates p by $-\partial_s(dG_s)$, so it carries $(q, dG_{s_0}(q))$ to $(q, dG_{s_1}(q))$. Compact support ensures completeness on the region of interest. □

5 Proof of Theorem 1

Step 1: local smoothing near each vertex

Let v be a vertex of K. Translate so $v = 0$ and restrict to a small closed ball B_v meeting no other vertex and intersecting K in the star of v. By Proposition 2, after a linear symplectomorphism we may assume

$$K \cap B_v = \mathrm{graph}(dF_v) \cap B_v$$

for a C^1 piecewise quadratic F_v with four sectors.

Choose $\delta_v > 0$ so that $B_{2\delta_v} \subset \pi(B_v)$ in base coordinates, and define the smoothed potential $\widetilde{F}_{v,t} := \widetilde{F}_{t,\delta_v}$ using mollification scale $\varepsilon = t$ and cutoff scale δ_v. Then for each $t > 0$, $\widetilde{F}_{v,t}$ is smooth and equals F_v outside $B_{2\delta_v}$. Hence

$$K_t^{(v)} \cap B_v := \mathrm{graph}(d\widetilde{F}_{v,t}) \cap B_v$$

is a smooth Lagrangian surface patch that agrees with K near ∂B_v and converges to $K \cap B_v$ in C^0 as $t \to 0$ (indeed in C^1 away from the vertex).

Because the deformation is by varying a generating function with compact support, Proposition 3 shows that for fixed v the family $K_t^{(v)}$ $(t > 0)$ is a Hamiltonian isotopy, supported in B_v.

Step 2: local smoothing along each edge away from its endpoints

After smoothing all vertices (in pairwise disjoint balls), the remaining singular set is a finite union of open edge segments where two faces meet.

Let e be an edge, and let U_e be a tubular neighborhood of a compact subsegment of e disjoint from the vertex balls. On U_e the surface is locally the union of exactly two smooth Lagrangian half-planes meeting along the edge line. As in Proposition 1, after a (local) symplectic linear change of coordinates we may write this as the graph of dF_e for a C^1 function F_e which is quadratic on the two sides of a line in the base. Mollify transversely and apply a cutoff in U_e exactly as above to obtain a family of smooth potentials $\widetilde{F}_{e,t}$ which agree with F_e near ∂U_e.

Again, by Proposition 3, this smoothing along e is realized by a Hamiltonian isotopy supported in U_e.

Step 3: assemble the local Hamiltonians

Because K is a finite polyhedral complex, there are finitely many vertices and edges. Choose:

- closed balls B_v around each vertex v with pairwise disjoint interiors, and
- closed edge neighborhoods U_e for each edge e, disjoint from all B_v and pairwise disjoint.

This is possible since the singular locus (1–skeleton) is compact and finite.

For each $t > 0$, perform the vertex smoothings inside each B_v and the edge smoothings inside each U_e, leaving the surface unchanged elsewhere. Since these neighborhoods are disjoint, the corresponding Hamiltonian isotopies have disjoint supports, and hence may be composed (or, equivalently, their generating Hamiltonians may be summed) to obtain a global Hamiltonian isotopy of smooth Lagrangian surfaces.

Denote the resulting smooth Lagrangian surface by K_t.

Step 4: extension to $t = 0$ as a topological isotopy

By construction, for each $t > 0$ the surface K_t agrees with K outside the union of the neighborhoods. Inside each neighborhood, K_t is a graph of $d\widetilde{F}_{(\cdot),t}$ and converges in C^0 to the original piecewise–linear (piecewise quadratic in the graph coordinates) model as $t \to 0$ by Lemma 3 and the cutoff construction.

Define a map $\Psi_t : K \to \mathbb{R}^4$ by:

- Ψ_t is the identity on K outside the smoothing neighborhoods, and
- in each neighborhood, use the base projection chart in which K is graph(dF) and set

$$\Psi_t(q, dF(q)) := (q, d\widetilde{F}_t(q)).$$

These local definitions are compatible on overlaps because the smoothing neighborhoods are disjoint and because the cutoff makes $\widetilde{F}_t = F$ near the boundary of each neighborhood.

Then Ψ_t is continuous for each $t \in [0, 1]$, is an embedding for $t > 0$, and $\Psi_t(K) = K_t$ for $t > 0$ while $\Psi_0(K) = K$. Standard topological arguments (or the isotopy extension theorem applied on the complement of a finite union of balls, plus explicit control inside each ball) show that $\{\Psi_t\}$ is a topological isotopy of embedded surfaces.

Thus $\{K_t\}_{t\in(0,1]}$ is a Hamiltonian isotopy of smooth Lagrangian surfaces extending to $K_0 = K$, i.e. a Lagrangian smoothing.

This completes the proof of Theorem 1. □

6 Consequences and remarks

Remark 2 (A topological restriction)**.** If a polyhedral Lagrangian surface K with the hypotheses of Theorem 1 existed with $H^1(K; \mathbb{R}) = 0$ (e.g. $K \cong S^2$), then the smoothing would produce a smooth embedded Lagrangian with vanishing H^1, hence an *exact* Lagrangian (since $\lambda_0|_{K_t}$ is closed and $H^1 = 0$), which is impossible in $(\mathbb{R}^4, d\lambda_0)$ by Gromov's theorem on exact Lagrangians. Therefore, such polyhedral Lagrangian spheres cannot exist. Theorem 1 should thus be viewed as a smoothing/rigidity mechanism: combinatorially simple embedded polyhedral Lagrangians (with four faces at each vertex) must already lie in the smooth Hamiltonian world.

References

References

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